

JUNGMIN KIM, PH.D.

Senior Photonics Engineer | Computational Photonics, Inverse Design, and Optical Computing
Madison, WI | [in jmkim93](https://www.linkedin.com/in/jmkim93) | [github jmkim93.github.io](https://github.com/jmkim93) | [✉ jmkim93@gmail.com](mailto:jmkim93@gmail.com) | [☎ +1 \(608\) 377-5049](tel:+16083775049)

SUMMARY

Photonics researcher and engineer with extensive experience in electromagnetic simulation, modeling, and inverse design. Skilled in Python-based automation and physics-driven optimization for complex optical systems.

EDUCATION

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|-------|---|----------------------------------|
| Ph.D. | Electrical and Computer Engineering
<i>Seoul National University</i>
Advisors: Prof. Namkyoo Park and Prof. Sunkyu Yu | Mar. 2017–Feb. 2023
Seoul, KR |
| B.S. | Electrical and Computer Engineering
<i>Seoul National University</i> | Mar. 2012–Feb. 2017
Seoul, KR |

WORK EXPERIENCE

- Flexcompute Inc.**, *Senior Photonics Engineer*, Madison, WI May 2026–Present
- Develop and validate Tidy3D workflows for integrated photonic device simulation, optimization, and fabrication-aware design.
 - Advise customers on simulation strategies, optimization workflows, and device design.
 - Create technical documentation, tutorials, and reproducible examples.
- University of Wisconsin–Madison**, *Research Associate, Scientist I*, Madison, WI Apr. 2023–Apr. 2026
- Mentor: Prof. Zongfu Yu
 - Designed a mmWave face verification system using metasurfaces, achieving > 90% accuracy.
 - Designed an incoherent meta-imaging system for low-light environments, with an approximately 2.9 dB PSNR enhancement.
 - Designed a photonic systolic array for all-optical matrix–matrix multiplication, achieving a theoretical computing density of $4.4 \text{ PMACS mm}^{-2} \text{ s}^{-1}$.
- Seoul National University**, *Research/Teaching Assistant*, Seoul, KR Mar. 2017–Feb. 2023
- Designed tilted Dirac cones and investigated temporally disordered modulation in photonic systems to achieve target scattering responses.
 - Assisted with ECE courses in Quantum Mechanics, Electromagnetism, and Nanophotonics.

CORE EXPERTISE

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| Simulation | FDTD (Tidy3D), FEM (COMSOL), RCWA, plane-wave expansion method (PWEM) |
| Programming & ML | Python, MATLAB, PyTorch |
| Optimization | Adjoint methods, automatic differentiation, gradient-based inverse design |
| Technical domains | Integrated photonics, metasurfaces, optical computing, photonic band structures, computational imaging, disordered photonics |

PUBLICATIONS

- [1] Q. Zhou^{†*}, **J. Kim**[†], Y. Tao[†], et al., “Quantum nonlinearity for optical neural computing”, [Nature Communications](#), in press, DOI:10.1038/s41467-026-77202-y (2026).

- [2] D. Lee[†], J. Yeo[†], G. Lee, et al., “Tailoring defects in photonic time crystals for coherent energy control”, [arXiv, 2605.30633 \(2026\)](#).
- [3] **J. Kim**^{†*}, Q. Zhou^{†*}, and Z. Yu, “Photonic systolic array for all-optical matrix–matrix multiplication”, [Laser & Photonics Reviews](#) **20**, e01995 (2026).
- [4] **J. Kim**^{*}, N. Yu, and Z. Yu, “Compute-first optical detection for noise-resilient visual perception”, [ACS Photonics](#) **12**, 1137–1145 (2025).
- [5] J. Kim[†], J.-Y. Kim[†], **J. Kim**, et al., “Inverse design of nanophotonic devices enabled by optimization algorithms and deep learning: recent achievements and future prospects”, [Nanophotonics](#) **14**, 121–151 (2025).
- [6] **J. Kim**, D. Lee, S. Yu^{*}, and N. Park^{*}, “Unidirectional scattering with spatial homogeneity using correlated photonic time disorder”, [Nature Physics](#) **19**, 726–732 (2023).
- [7] S. Choi[†], **J. Kim**[†], J. Kwak, N. Park^{*}, and S. Yu^{*}, “Topologically protected all-optical memory”, [Advanced Electronic Materials](#) **8**, 2200579 (2022).
- [8] S. Oh, **J. Kim**, X. Piao, et al., “Control of localization and optical properties with deep-subwavelength engineered disorder”, [Optics Express](#) **30**, 28301–28311 (2022).
- [9] S. Park[†], I. Lee[†], **J. Kim**, N. Park^{*}, and S. Yu^{*}, “Hearing the shape of a drum for light: isospectrality in photonics”, [Nanophotonics](#) **11**, 2763–2778 (2022).
- [10] **J. Kim**, S. Park, S. Yu^{*}, and N. Park^{*}, “Machine-engineered active disorder for digital photonics”, [Advanced Optical Materials](#) **10**, 2102642 (2022).
- [11] **J. Kim**, S. Yu^{*}, and N. Park^{*}, “Universal design platform for an extended class of photonic dirac cones”, [Physical Review Applied](#) **13**, 044015 (2020).

[†] Co-first authors; ^{*} Corresponding author(s)

PRESENTATIONS

- [C1] Q. Zhou, J. Kim, Y. Tao, G. Huang, and Z. Yu, “Low-power optical neural networks enabled by inverse-designed quantum nonlinearity”, in [CLEO 2026](#) (2026), SM3E.3.
- [C2] J. Kim, N. Yu, and Z. Yu, “Incoherent meta-imaging system for noise-robust object recognition”, in [2024 conference on lasers and electro-optics pacific rim \(CLEO-PR\)](#) (2024), Mo1A4.
- [C3] D. Lee, J. Kim, H. Park, et al., “Design of correlated photonic time disorder for unidirectional scattering”, in [Advanced photonics congress 2023](#) (2023), NoTu3C.4.
- [C4] J. Kim, S. Park, D. Lee, S. Yu, and N. Park, “Data-driven engineering of active photonic disorder”, in [Frontiers in optics + laser science 2022 \(FIO+LS\)](#) (2022), JW4A.20.
- [C5] D. Lee, J. Kim, N. Park, and S. Yu, “Molecular dynamics for microscopic analysis of refractive index in amorphous hafnium oxides”, in [Frontiers in optics + laser science 2022 \(FIO+LS\)](#) (2022), FW7C.5.
- [C6] J. Kim, S. Yu, and N. Park, “Neural-network-based design of tunable multilayer films”, in [OSA advanced photonics congress 2021](#) (2021), JW4B.3.

HONORS AND AWARDS

Feb. 2023 Distinguished Dissertation Award, ECE, Seoul National University
 Feb. 2019 Best Poster Award, OSK Winter Meeting
 2014–2016 National Science & Engineering Undergraduate Scholarship, Korea Student Aid Foundation

ACADEMIC SERVICE

Reviewer *Optica, Photonics Research, Optics Express, Optics Letters*
 Nature Electronics, Nature Communications
 ACS Photonics, Advanced Functional Materials